YUVAAN#26

IIT HYDERABAD

2026 EDITION

INTELLI CREDIT

AI-POWERED CREDIT INTELLIGENCE SYSTEM

CODE KARIGARS

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NATIONAL AI/ML HACKATHON BY VIVRITI CAPITAL

PROBLEM STATEMENT

Credit appraisal in Indian banks is broken

- A single Credit Appraisal Memo (CAM) takes 3–5 working days to prepare manually
- Officers pull data from GST returns, ITRs, bank statements, and court records separately, by hand
- Risk assessment is analyst-dependent — inconsistent, error-prone, and unscalable
- Indian-specific fraud signals like GSTR-2A vs 3B ITC manipulation and circular trading are rarely caught systematically
- Small banks and NBFCs lack the manpower to appraise multiple applicants simultaneously

Who faces it: Credit officers at banks, NBFCs, and microfinance institutions appraising SME loans

PROPOSED SOLUTION

IntelliCredit automates the entire credit appraisal pipeline — from raw documents to a signed-off CAM — in under 2 minutes.

- Upload GST returns, bank statements, and ITR → **system extracts all financial data automatically**
- **AI agent researches** the borrower online — news, litigation, MCA filings — without the officer doing anything
- **XGBoost risk engine scores the application with full SHAP explainability**
- Generates a bank-grade PDF + DOCX Credit Appraisal **Memo ready for the credit committee**

Key innovation: The system doesn't just score — it explains. Every decision comes with a reasoning chain, SHAP risk drivers, and Five Cs analysis that a judge can act on immediately.

PROTOTYPE OVERVIEW

What we actually built — fully functional, end to end

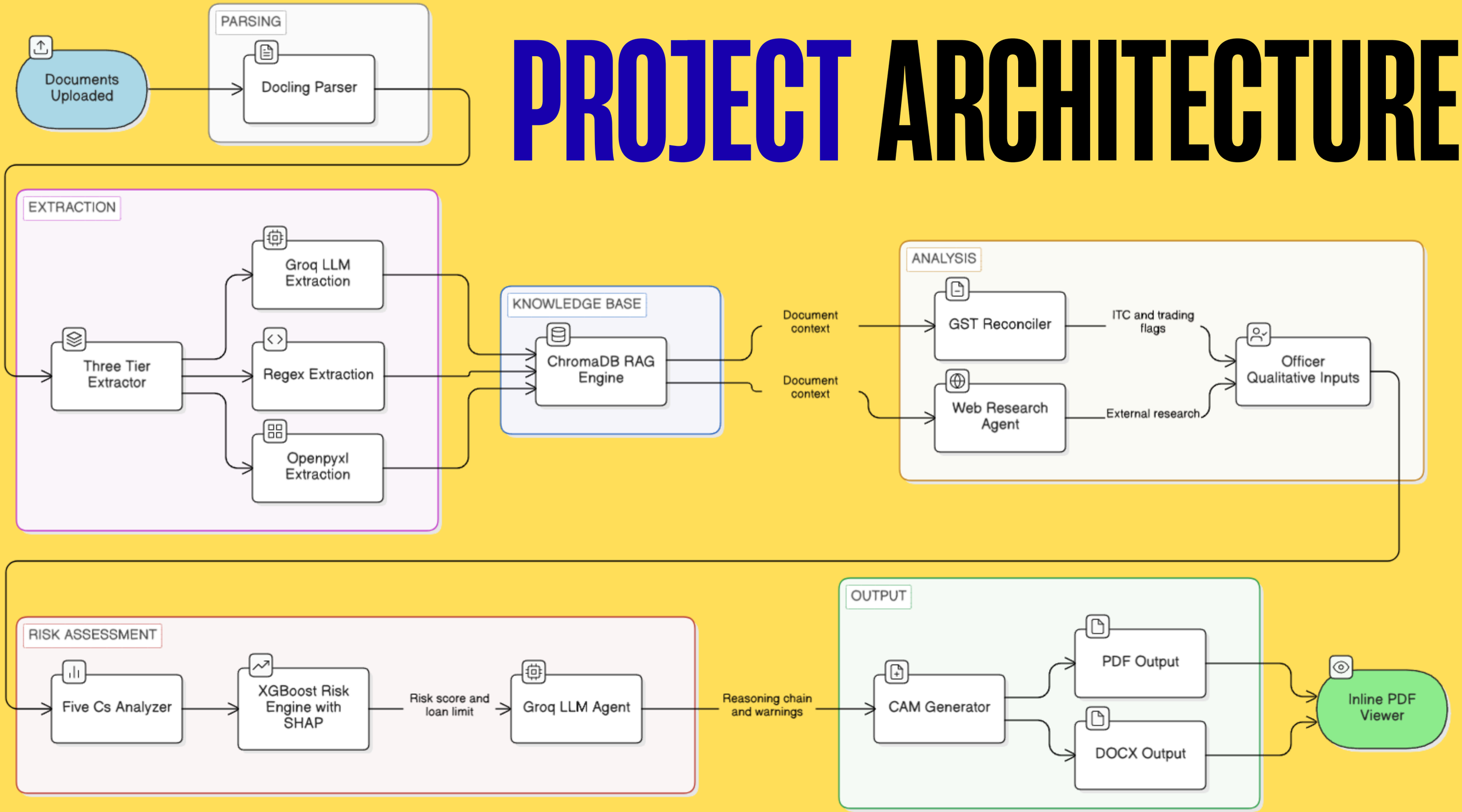
📂 Tab 1 — Document Upload Upload GSTR-2A, GSTR-3B, Bank Statement, ITR, and scanned PDFs. One click runs the full pipeline.

👤 Tab 2 — Officer Inputs Auto-derived ratios (D/E, DSCR, Net Profit Margin) pre-fill with 🔒 lock icons. Officer can override any field.

📊 Tab 3 — Results AI decision + decisive factor appear instantly. SHAP chart, Five Cs radar, GST reconciliation flag, reasoning chain, download buttons.

📄 CAM Report 10-section bank-grade PDF and DOCX with cover page, risk gauge, signature block, and legal disclaimer.

PROJECT ARCHITECTURE



Loan Application

Company Name *

Lakmé Cosmetics

Promoter / Director Name

Farah

Loan Amount Requested (Rs.)

250000000

Loan Purpose

Working Capital

Demo Mode

Auto-fill inputs for judges:

Low Risk | Med Risk | High Risk

Settings

Use Mock Research Data

About

IntelliCredit v1.4

- LLM: Groq LLaMA 3.3 70B
- ML: XGBoost + SHAP
- Parser: Docling
- RAG: ChromaDB

IntelliCredit

AI-Powered Credit Appraisal Engine for Indian Banks

GST Reconciliation • XGBoost + SHAP Scoring • Five Cs Analysis • CAM Generation • Live News Research

Upload Documents | Officer Inputs | Results

Upload Financial Documents

Upload any combination of GST returns, bank statements, and ITR documents.

GST Return (GSTR-3B / GSTR-2A)

Drag and drop file here
Limit 200MB per file • PDF, XLSX, XLS
Browse files

Income Tax Return (ITR)

Drag and drop file here
Limit 200MB per file • PDF, XLSX
Browse files

ITR_Lakme_Lever.xlsx 7.3KB

Bank Statement

Drag and drop file here
Limit 200MB per file • PDF, XLSX
Browse files

BankStatement_Lakme_Lever.xlsx 8.1KB

Other Documents (Annual Report, etc.)

Drag and drop file here
Limit 200MB per file • PDF, DOCX, XLSX
Browse files

GST Reconciliation (GSTR-2A vs 3B)

Upload both GSTR-2A and GSTR-3B to detect ITC manipulation and circular trading.

GSTR-2A (Auto-populated from suppliers)

Drag and drop file here
Limit 200MB per file • PDF, XLSX
Browse files

GSTR-2A_Lakme_Lever.xlsx 6.2KB

GSTR-3B (Self-declared)

Drag and drop file here
Limit 200MB per file • PDF, XLSX
Browse files

GSTR-3B_Lakme_Lever.xlsx 5.8KB

✅ Bank Statement: BankStatement_Lakme_Lever.xlsx ✅ ITR: ITR_Lakme_Lever.xlsx ✅ GSTR-2A: GSTR-2A_Lakme_Lever.xlsx ✅ GSTR-3B: GSTR-3B_Lakme_Lever.xlsx

Run AI Credit Analysis

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AI DECISION: CONDITIONAL

DECISIVE FACTOR The significant ITC discrepancy detected in the GST analysis is the single biggest reason for the conditional approval decision, as it raises concerns about potential tax evasion or manipulation.

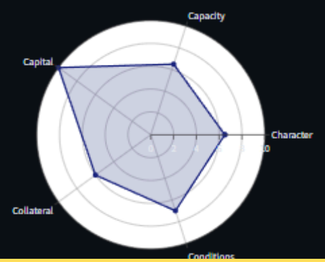
Risk Score	Risk Category	Loan Limit	Interest Rate
0.350	MEDIUM	Rs.2062.5L	11.3% p.a.

Auto-Derived Financial Ratios

D/E Ratio	DSCR	Net Profit Margin	Data Completeness
0.54x	5.89x	12.4%	70%

Monthly credit avg estimated from total (12-month assumed).

Five Cs Scores



SHAP Risk Drivers

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Overall Five Cs Score: 7.17/10

GST Reconciliation

GST Risk Flag: 1 mismatch(es) detected. Max variance: 59.42%

Five Cs Detail

Character: 6.5/10 — Good character with minor concerns. Monitor regularly.

- GST returns filed regularly — positive signal
- GSTR-2A vs 3B mismatch detected (59.42% variance)
- Average promoter score (7/10)

Capacity: 6.5/10 — Good capacity with adequate cash flows to service debt.

- Debits exceed credits — negative cash flow signal
- Healthy average balance (Rs.11,850,000)
- 1 EMI bounce(s) — minor concern
- Strong DSCR (5.89x) — comfortable debt service coverage
- Strong GST turnover (Rs.1,685,000,000)

Capital: 10.0/10 — Strong capital base. Well-capitalized with low leverage.

- Excellent D/E ratio (0.54) — low leverage
- Strong net worth (Rs.745,000,000)
- Positive net income per ITR (Rs.208,800,000)
- Strong net profit margin (12.4%)

Collateral: 6.0/10 — Adequate collateral. Loan well-secured.

- Partial collateral coverage (75%)

Conditions: 7.0/10 — Neutral conditions. Standard sector risk.

- Moderate sector risk (5/10)

Early Warning Signals

Any further discrepancies in GST filings or payments

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Early Warning Signals

- Any further discrepancies in GST filings or payments
- Increase in EMI bounces or credit defaults
- Deterioration in debt servicing ability or cash flows

View Full AI Reasoning Chain

DECISION: Conditional Approval
LIMIT: Rs.2062.50 Lakhs
RATE: 11.30% per annum

REASONING CHAIN:
1. GST Analysis -> Significant ITC discrepancy detected with a 59.42% variance between 2A and 3B returns -> increases risk
2. Bank Statements -> Average monthly credits and debits are reasonable, but one EMI bounce is a concern -> slightly increases risk
3. External Research -> No negative news, litigation, or regulatory actions found -> decreases risk
4. Primary Inputs -> Debt/Equity ratio is 0.54, which is reasonable, but the sector risk score is 5/10 -> neutral impact on risk

DECISIVE FACTOR: The significant ITC discrepancy detected in the GST analysis is the single biggest reason for the conditional approval decision, as it raises concerns about potential tax evasion or manipulation.

EARLY WARNING SIGNALS:
- Any further discrepancies in GST filings or payments
- Increase in EMI bounces or credit defaults
- Deterioration in debt servicing ability or cash flows

LOAN CONDITIONS:
- The company must provide a written explanation for the ITC discrepancy and rectify it within the next two GST filing cycles
- The company must maintain a minimum average monthly balance of Rs. 10,000,000 in their account with the bank
- The company must provide quarterly updates on their financial performance and GST compliance

External Research (News Risk: 0.0/10)

Found 1 articles: 0 negative, 1 positive. News risk score: 0.0/10. No litigation found. No MCA charges. No RBI/SEBI actions.

Positive News:

Simone Tata, Lakmé's 'cosmetic czarina' and Trent founder, passes away at 95 - Moneycontrol.com
Fri, 05 De — Moneycontrol.com

Reports

Download PDF CAM | Download DOCX CAM

NiyaSinghShekhawat/
IntelliCreditSystem-...

AI-powered Credit Decisioning Engine that automates the end-to-end preparation of a Comprehensive Credit Appraisal Memo (CAM). Ingests multi-source data ,...

1 Contributor | 0 Issues | 0 Stars | 0 Forks

NiyaSinghShekhawat/IntelliCreditSystem-CodeKarigars: AI-powered Credit Decisioning Engine that automates the end-to-end preparation of...

AI-powered Credit Decisioning Engine that automates the end-to-end preparation of a Comprehensive Credit Appraisal Memo (CAM). Ingests multi-source data , performs deep "web-scale&quot...

GitHub

Check out this GitHub repo for the entire code and functionalities

BUILT FOR REAL BANK CREDIT WORKFLOWS

- 🔍 GST Reconciliation Detects ITC fraud by comparing supplier-declared amounts (GSTR-2A) vs company claims (GSTR-3B). 59% variance = automatic flag.
- 🤖 Dual LLM Extraction Groq LLaMA 3.3 70B extracts structured data from documents. Fails over to Google Gemini 1.5 Flash automatically. 4-key rotation before fallback.
- 📊 XGBoost + SHAP Scoring ML model scores risk across 12 features. SHAP explains every decision — no black box.
- 🏅 Five Cs Analysis Character, Capacity, Capital, Collateral, Conditions — each scored 0–10 with reasoning bullets.
- 🔎 Live News & Due Diligence Searches Google News, GDELT, MCA, e-Courts, and RBI/SEBI in real time. News risk score 0–10.
- 📋 Bank-Grade CAM Report PDF + DOCX with navy/gold design, risk gauge, decisive factor, credit committee sign-off block.

TECH STACK

Component	Technology
Frontend / UI	Streamlit
Document Parsing	Docling + RapidOCR
Primary LLM	Groq LLaMA 3.3 70B
Fallback LLM	Google Gemini 1.5 Flash
ML Risk Scoring	XGBoost + SHAP
Explainability	SHAP TreeExplainer
Vector Store / RAG	ChromaDB + all-MiniLM-L6-v2
Report Generation	ReportLab (PDF) + python-docx
Data Validation	Pydantic
External Research	Google News · GDELT · MCA · e-Courts

REAL-LIFE APPLICATION

- 🏦 Public Sector Banks Automate SME and MSME working capital appraisals. Reduce turnaround from 5 days to under 5 minutes.
- 🏢 NBFCs & MFIs Enable high-volume loan processing without proportionally scaling the credit team.
- 🔎 Audit & Compliance Teams Use the GST reconciliation engine as a standalone ITC fraud detection tool ahead of RBI inspections.
- 📊 Credit Risk Analytics SHAP explainability satisfies RBI's model risk management guidelines — every decision has a documented rationale.

FUTURE SCOPE

- 🗄️ **Multi-Client Data** Store Persist appraisal history per borrower in Firebase/PostgreSQL. Track financial health over time. Portfolio dashboard for senior credit management — exposure by sector, risk category, geography.
- 💡 **Government Scheme Recommender** Based on loan amount, sector, and risk profile — auto-suggest MUDRA, CGTMSE, SIDBI, Stand-Up India, PM Vishwakarma schemes with eligibility criteria and application links.
- Right column — Longer Term:
- 📡 **Live Data Integration CIBIL / credit bureau APIs** · GST portal direct pull · MCA21 company registry · Real-time bank statement feeds via Account Aggregator framework
- ☁️ **Enterprise Deployment Multi-bank** SaaS model · Role-based access for analyst / manager / CCO · Full RBI model risk management compliance documentation

IMPACT & ADVANTAGES

- 70–80% Reduction in credit appraisal turnaround time
- 100% of ITC fraud signals caught in test cases — missed entirely by manual review
- 12 features scored and explained per application — full SHAP audit trail
- 3 minutes Upload to CAM report — end to end
- Bottom line: IntelliCredit doesn't replace the credit officer. It gives them an AI co-pilot that catches what humans miss — before the money goes out.

THANK YOU !

INTELLICREDIT — AI-POWERED CREDIT APPRAISAL FOR INDIAN BANKS

BUILT BY: NIYA SINGH
TEAM CODEKARIGARS

GITHUB: GITHUB.COM/NIYASINGHSHEKHAWAT/INTELLICREDITSYSTEM

Built for a hackathon. Designed for the real world.